

Development of a 3D-Printed Educational Tool for Visualization of Amino Acid Side Chain Effects on Protein Folding

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INTRODUCTION

“Structure determines function” is a fundamental theme in biology. Thus, it is essential for students to understand how the properties of amino acids affect the higher-order structures in proteins (Figure 1). The chemical properties of the amino acid side chains which are divided into four groups: nonpolar, polar, acidic, and basic (Figure 2) significantly influence the formation of the secondary and tertiary structures of a protein.

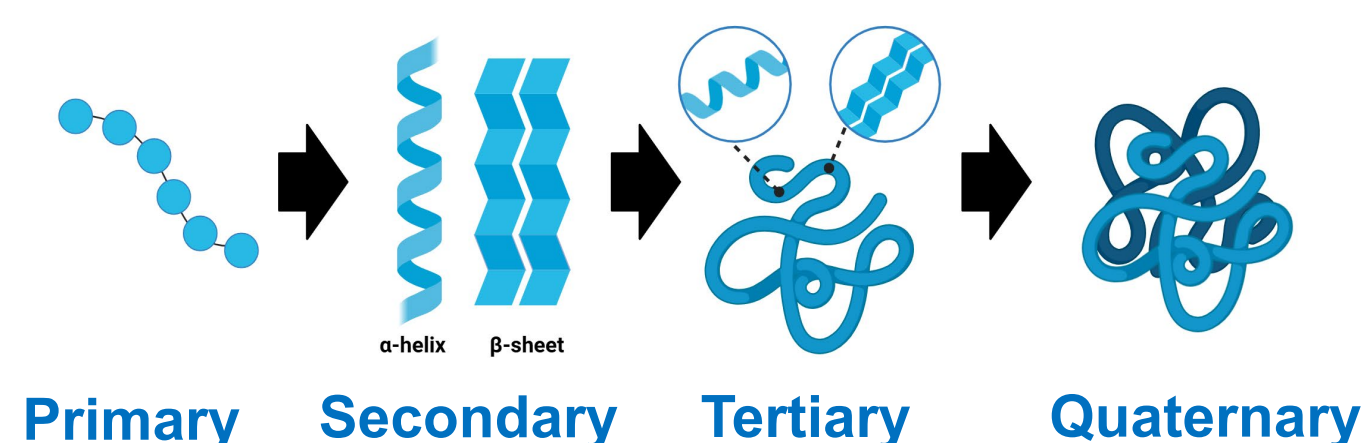


Figure 1. Hierarchy of protein structure where amino acid sequence determines higher order structures.

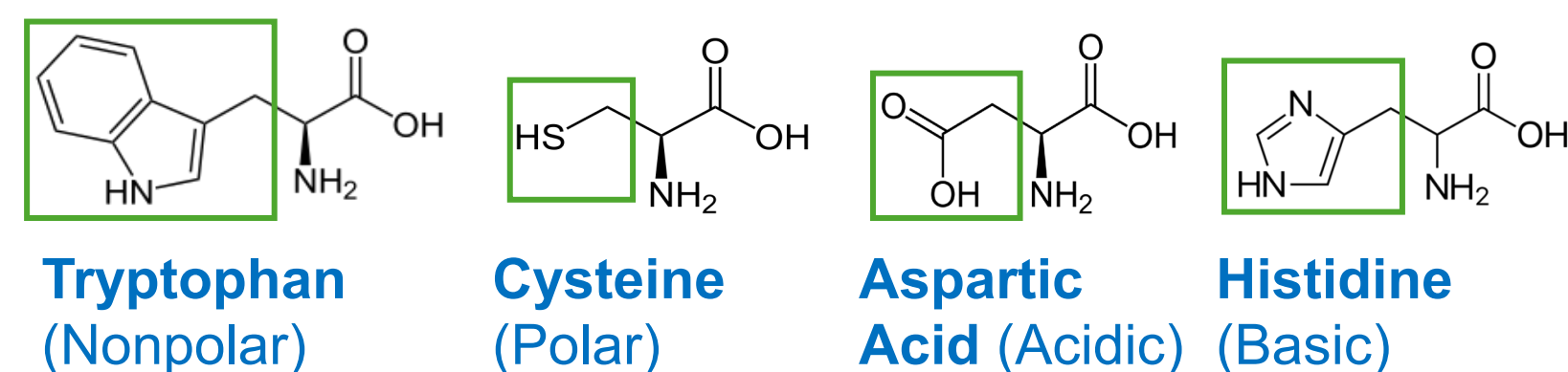


Figure 2. Examples of amino acid side chains targeted in green.

Although there exist educational tools to demonstrate protein folding, it is difficult to represent its dynamic nature using static 2D methods. Furthermore, existing computer simulations lack the benefits of hands-on tools that allow students to build strong spatial intuition. Most available physical models focus on the polypeptide backbone and overall protein fold rather than the side chains. Therefore, there is a need for a 3D model that is both scientifically meaningful and classroom-friendly, to accurately represents the effects of amino acid chemistry on protein folding. The objective of this project is to develop a 3D-printed modular protein-folding model that accurately represents amino acid structures. In addition, providing a tangible, manipulable tool to enhance spatial understanding through hands-on learning, 3D printing offers an affordable and customizable way that is sustainable for small schools with limited budget.

DESIGN CRITERIA

A commercially available bendy tube with a 10 mm diameter was used to represent the protein backbone. This model was designed to work in conjunction with this bendy tube based on the following criteria:

- clearly representation of side chain structure and chemicals,
- modular and removable components for easy adaptation to different lessons and examples,
- durable,
- intuitive to use.

MODEL DESIGN

MODELING OF AMINO ACID SIDE CHAINS

ChemDraw was used to model the amino acids side chains in 2D. Chem3D was used to model the amino acids side chains in 3D and minimize their energy. UCSF Chimera was used to set the visualizing style of the side chains. Blender and MolPrint was used to remove hydrogens from the α -carbon on the side chains and add double bonds. The scheme of the workflow is shown in Figure 3.

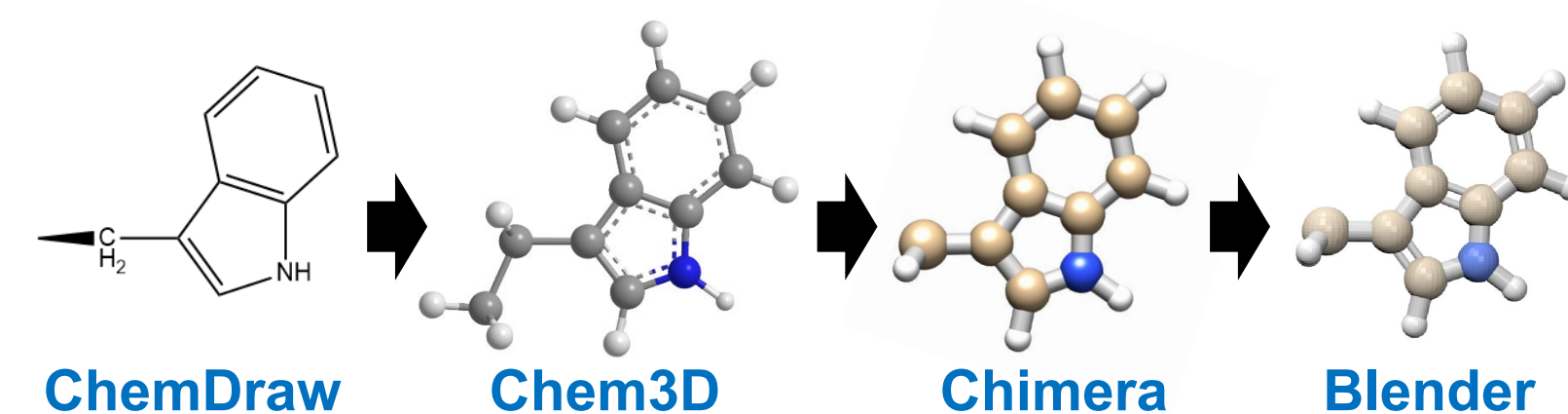


Figure 3. Sample workflow of amino acid modeling.

ATTACH MECHANISM DESIGN

Figure 4 shows the attachment mechanism which was designed in two parts, the **clamp** attaches to the bendy tube, and the **joint**, which attaches to amino acid at the α -carbon. Many iterations of both designs were tested to improve the balance of rigidity and flexibility. To provide wide adaptability, α -carbons common to all amino acids were used as the target for the joint to side chains. The final joint design secured the bond from the α -carbon but provided room for rotation; giving the side chains more freedom to orient themselves as they do reality. The final iteration of the clamp makes tight contact with the bendy tube yet allowed for movements around as well as up and down the tube. Both parts were tested with repeated attachment, detachment and reattachment before combined and printed as one object.



Figure 4. Final design of the clamp and joint used for printing.

3D PRINTING AND FABRICATION

All components were 3D-printed in polylactic acid (PLA) filament on a Bambu Lab X-1 Carbon 3D-printer. Each side chain group was printed using a different filament color:

- **nonpolar** side chains in **purple**,
- **polar** side chains in **orange**,
- **acidic** side chains in **yellow**, and
- **basic** side chains in **green**.

Noncarbon atoms were painted on the side chains in their usual colors; hydrogen atoms in white, oxygen in red, nitrogen in blue, and sulfur in yellow. All side chains were sprayed painted with a topcoat to protect the paints. Attachment mechanisms were printed in grey to match the color of the bendy tube used.

RESULTS

Figure 5 shows a close-up of a 3D printed aspartic acid side chain attached to the bendy tube using the attachment fabricated based on the final camp-joint design. It can be seen that the joint connects to the side chain at the α -carbon around the C_{α} -C single bond. The larger circular ring provides room around the α -carbon to allow freedom of rotation. Oxygen atoms in the side chain is painted red and hydrogen painted white to make the intermolecular forces between side chains obvious when demonstrating their effect on higher order protein structure.



Figure 5. A close-up of 3D printed amino acid side chain, attachment and bendy tube assembly.

Multiple amino acid side chains can be attached to the bendy tube using the attachment printed to represent specific peptide sequence (Figure 6). The clamp can move around and along the bendy tube while the side chains can also rotated along the axis of the C_{α} -C single bond to fine tune the interactions between the amino acids. These the bending of the tube and the adjustments of side chains' spacing and orientations, interactions between the side chains can be demonstrated. The relationship to the side chain chemistry and the formation of secondary protein structure like α -helix and β -sheets can be visualized.



Figure 6. Multiple amino acid side chains attached to the bendy tube to model specific peptide sequence.

Figure 7 shows the printed amino acid side chains on a display stand specifically designed and printed as part of this project. The joint used for α -carbon is used to on the stand to hold the side chains. This provides ease in the model's storage and transportation as well as an aesthetics that can make protein structures more engaging to students.

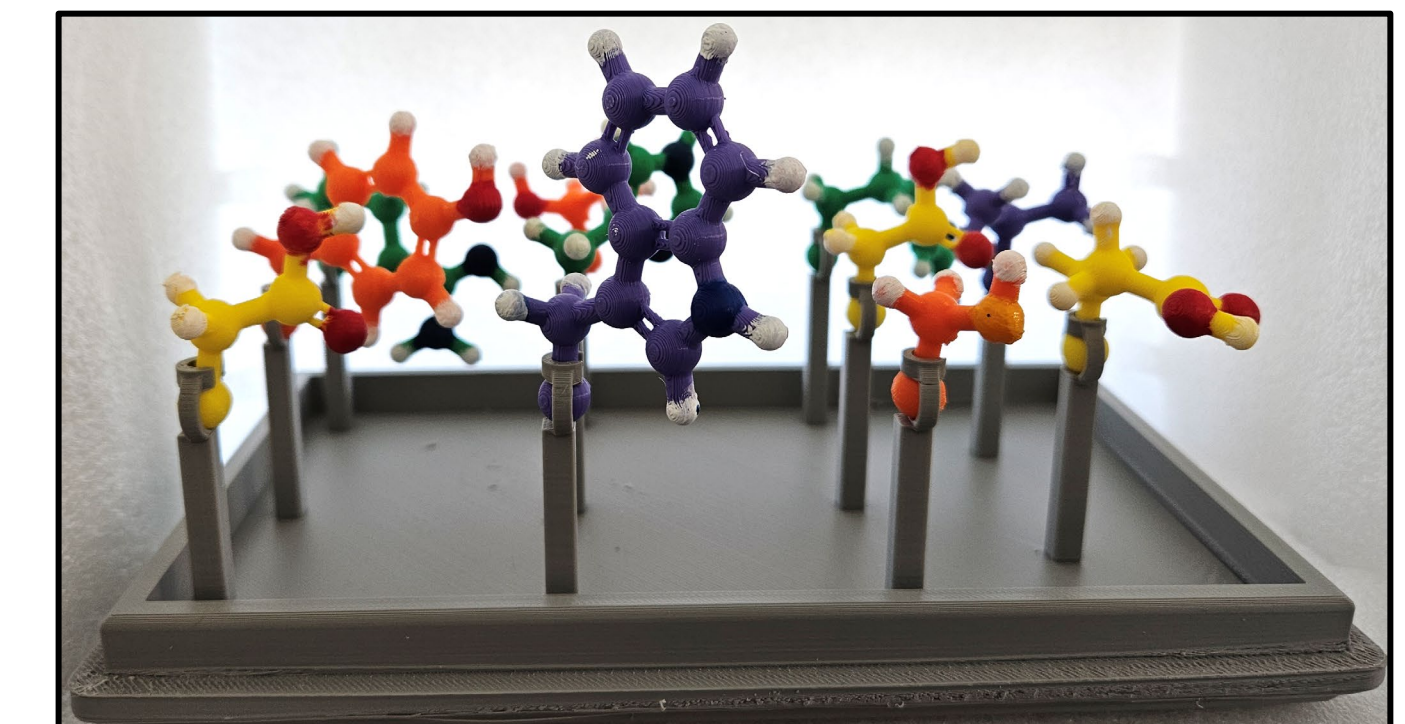


Figure 7. A display stand for the amino acid side chains.

CONCLUSION

This 3D-printed model was developed as a tactile pedagogical tool to assist instructors in teaching and students in learning the complexities of protein folding through bridging the gap between conceptual theory and experiential learning. This project demonstrated the potential of additive manufacturing in providing adaptable, affordable and impactful enhancement to improve biology and chemistry education. Based on this initial work, future iterations aim to incorporate magnetic attachments to represent intermolecular interactions such as hydrogen bonding. Furthermore, this framework could be expanded to a comprehensive biological modeling suite, simulating other fundamental processes such as DNA replication, transcription, translation, and operon gene regulation.

REFERENCES

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WEBSITE

This website provides the background and the development of this protein folding tool. Additionally, it includes the GitHub repository which hosts all STL files necessary to 3D-print the model.



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